

Supplementary Information B: treatment of fishing statistics data

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Data sources by fishing style

Commercial fishers

Data for fishery dependent information for professional commercial fishers are available through the European Data Collection Framework, which most recent update at the time of writing includes data until 2024. The dataset relevant for this study includes:

- *Fishery dependent information (FDI) for EU Fleets* (JRC, n.d.): provides fishing effort (dataset D1; fishing days) and catch (dataset D2; weight and value) data by country, area, fleet segment and target species
- *Annual Economic report on EU fishing fleet* (JRC 2025): provides economic overview (dataset D3; number of vessels, income, expenses) of european fishing fleets by country and fleet segment

The FDI data resolution allow to obtain information for the study area (Bothnian Sea - ICES area 27.3.D.30), while the AER is only available at country (Sweden) level. Different aggregation of the data requires to implement some simplifications or assumptions, hereby explained.

Small scale fishers

Small scale fishers catch and effort data correspond to vessels of size < 12m, deploying passive gears and operating in the sub-region 27.3.D.30. This region includes the majority of small scale coastal fishers deploying gillnets in Sweden, especially for those targeting Herring (more than 50% of total fishers with GNS-SPF metier are registered in the study area). Economic data for the fleet segments, although refers to values at country level, are considered to be representative of the study area.

Pelagic trawlers

Pelagic correspond to vessels of size > 24m operating pelagic nets (gear code: Trawl Midwater - TM) in the sub-region 27.3.D.30. Pelagic trawlers move between Swedish areas, and the same fleet can report catches in several divisions. Economic data at country level are therefore an appropriate aggregation. However, given the very small number of active vessels, Sweden does not provide economic data at fleet segment level for confidentiality reasons (JRC 2025). Economic data are provided for the aggregated Large Scale fleet, which includes vessels > 24m operating demersal trawlers, pelagic trawlers and purse seiners. On average, over the period 2020-2025 40 % of landed value.

Recreational fishers

Dataset for recreational fishers is available through Sweden statistic reports (SWAM 2024), which most recent update at the time of writing includes data until 2024. The dataset relevant for this study are the following tables (table number ios reported as it appears in Swam):

- Table 5: target species in the sea in 2024
- Table 6: primary and secondary gear used in recreational fishing in 2024
- Table 8: Catches in recreational fishing in the sea in 2024
- Table 9: Expenditures in recreational fishing in 2024, million SEK

Dataset ID	Dataset_name	Description
D1	fdi_effort_SWE.csv	Fishing effort for commercial fisheries from FDI data call (JRC, n.d.)
D2	fdi_catch_SWE.csv	Catches for commercial fisheries from FDI data call (JRC, n.d.)
D3	fdi_SWE.csv	Economic overview (number of vessels, income, expenses) of european fishing fleets (JRC 2025)

Dataset ID	Dataset_name	Description
D3	recf_catch.xlsx; tab6	Thousand of practitioners reported for recreational fishing technique in Sweden
D4	recf_catch.xlsx; tab2	Thousand of practitioners reported for target species in Sweden

Fishing gears

Small scale coastal fishers (ssf)

We analysed dataset D1, filtered to ICES area 27.3.D.30 and Swedish fleet, to identify the mostly used métiers by the ssf fishers (vessels categories VL0812 and VL0008), where a **métier** is defined as group of fishing activities targeting a similar species or assemblage of species, using similar gear. Figure C.1 shows average total fishing days by métier over the available data (years 2008-2024). We select for consideration in our study the métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). The sum of average fishing days for the three selected métiers accounts for 85% of total fishing days for ssf in our study area.

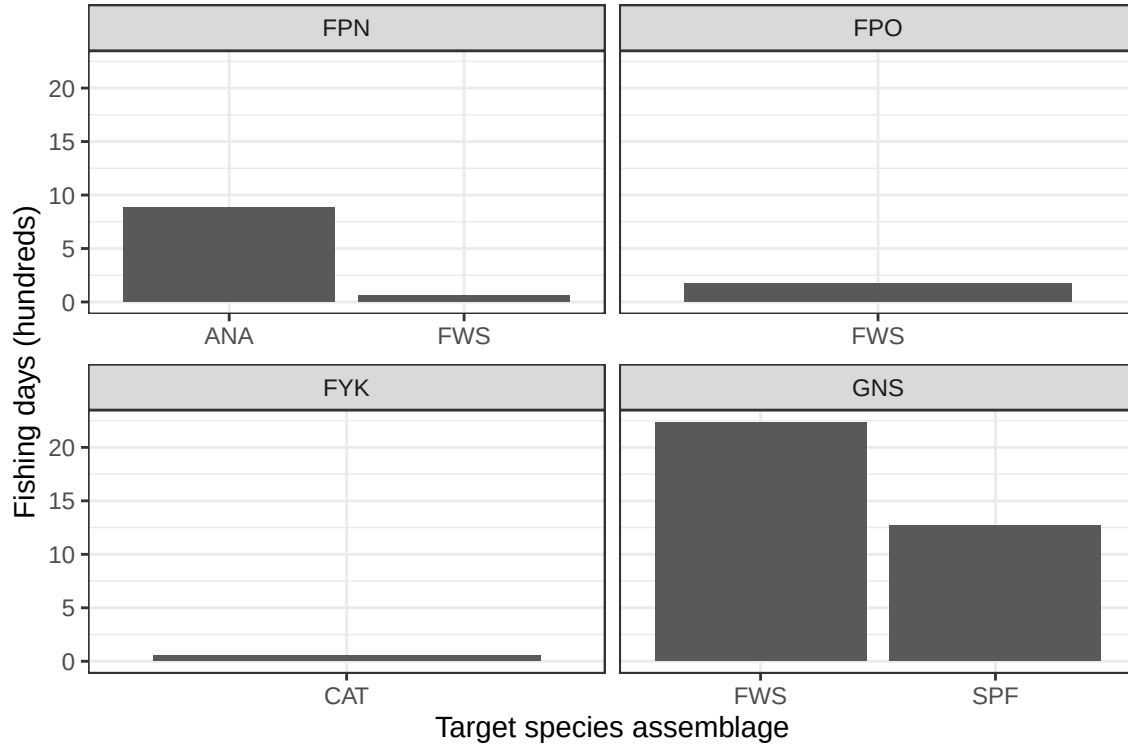


Figure C.1: Fishing days by metier, swedish small scale coastal fleet in Bothnian Sea

Pelagic trawlers (trw) and large scale fleet

We analysed dataset D1, filtered to Swedish fleet, to assess the fishing effort distribution by métier in the entire Swedish fleet. This information supports the interpretation of economic data.

Recreational fishers

Dataset D3 shows the number of practitioners for each recreational fishing method. Fishing methods based on hook and lines are dominating the distribution reported in Fig. C.3. Cages and nets are the two main techniques that do not base on hook and line. Cages are predominantly used in crustaceans fishery on the west coast of Sweden. The use of nets is popular in the Baltic Sea. For the purpose of this study we consider that only hook and line methods apply for the recreational fishing style, with no distinction among

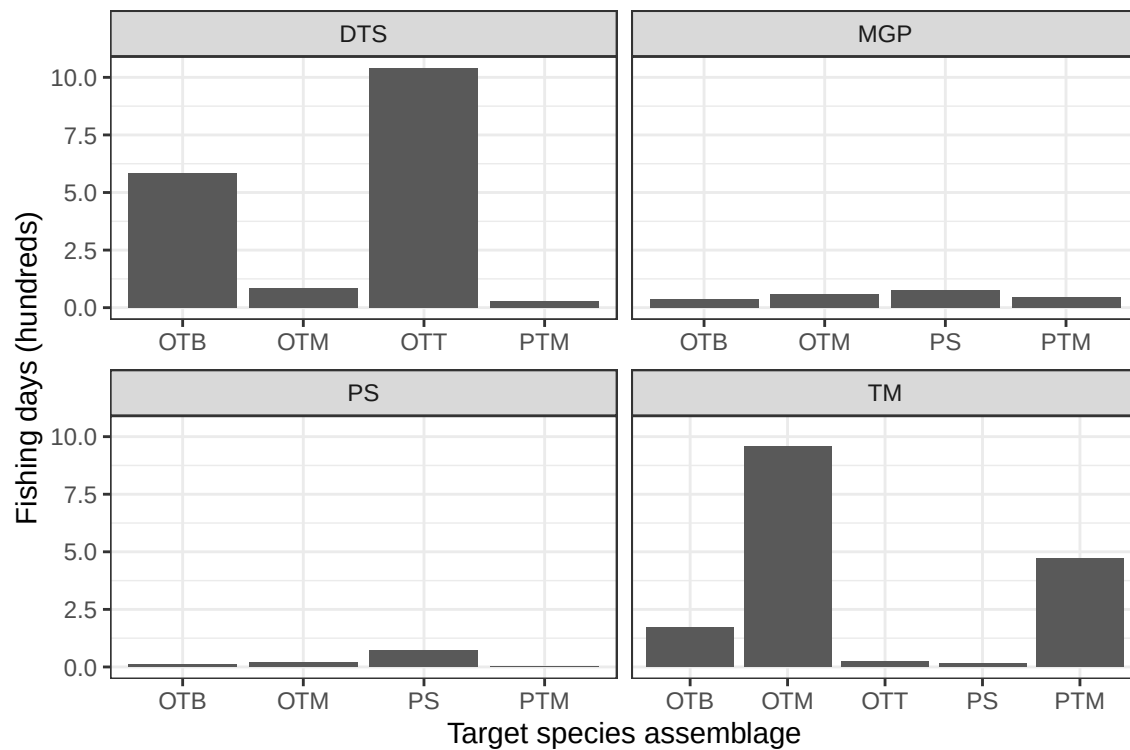


Figure C.2: Fishing days by metier, swedish large scale fleet

methods. We apply distinction between inland fishing and sea fishing, using estimates from Table 2 (5814 thousand practitioners inland; 4303 practitioners at sea). Nets applies to the household fishing style.

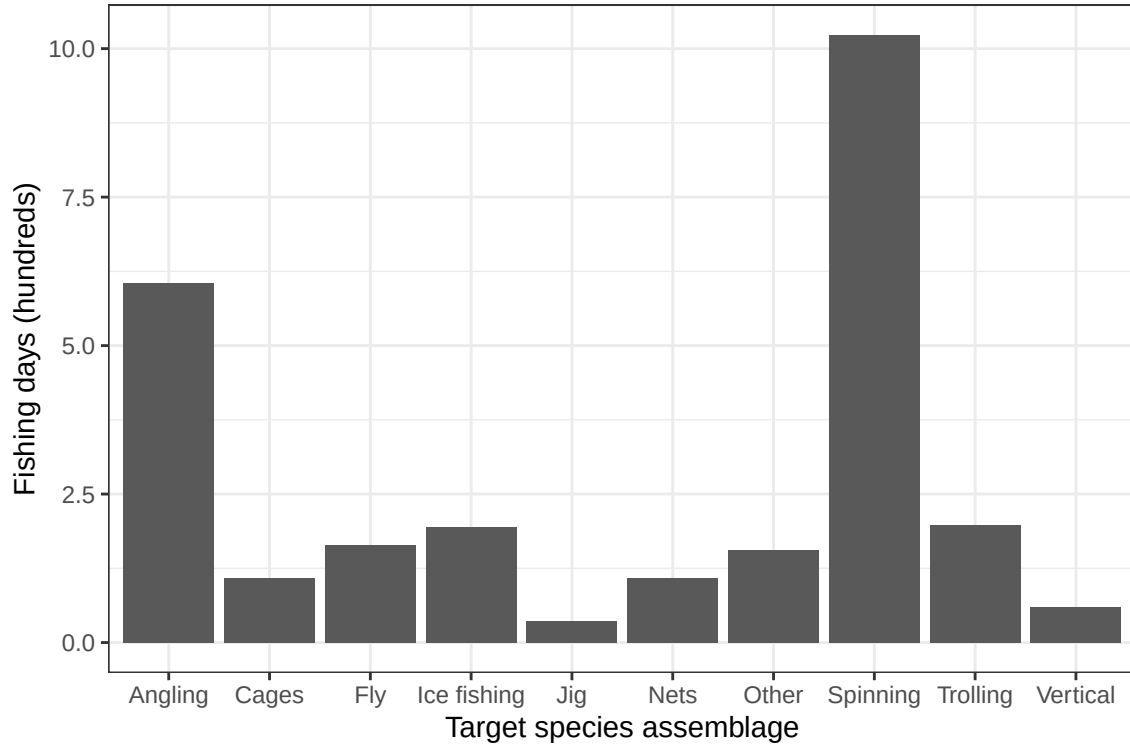


Figure C.3: Fishing days by fishing technique applied by swedish recreational fishers

Catches

Small scale coastal fishers

We analysed dataset D2 to identify the catch composition associated to ssf métiers identified in the previous section. The dataset, filtered to Swedish fleet in ICES area 27.3.D.30, vessels categories VL0812 and VL0008, was subset to match the ssf métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026). Catch data were aggregated by metier and averaged over years. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 95th percentile. Figure C.3 shows average tons by species and metier over the available data (years 2008-2024). The species mostly represented are Atlantic herring (target of gillnets for small pelagic fish), Atlantic salmon (main target of pound nets for anadromous species), European perch (main target of gillnets for freshwater species). Whitefish is also considered a relevant species as it significantly contributes to catches for FPN-ANA and GNS-FWS.

Pelagic trawlers

We analysed dataset D2 to identify the catch composition associated to pelagic trawlers. The dataset, filtered to Swedish fleet in ICES area 27.3.D.30, was subset to match the gear type “OTM”. Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026). Catch data were aggregated by metier and averaged over years. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 99th percentile. Figure C.4 shows that Atlantic herring contributes for more than 97% of average catches in the study area. For pelagic trawlers we only consider Herring as relevant species.

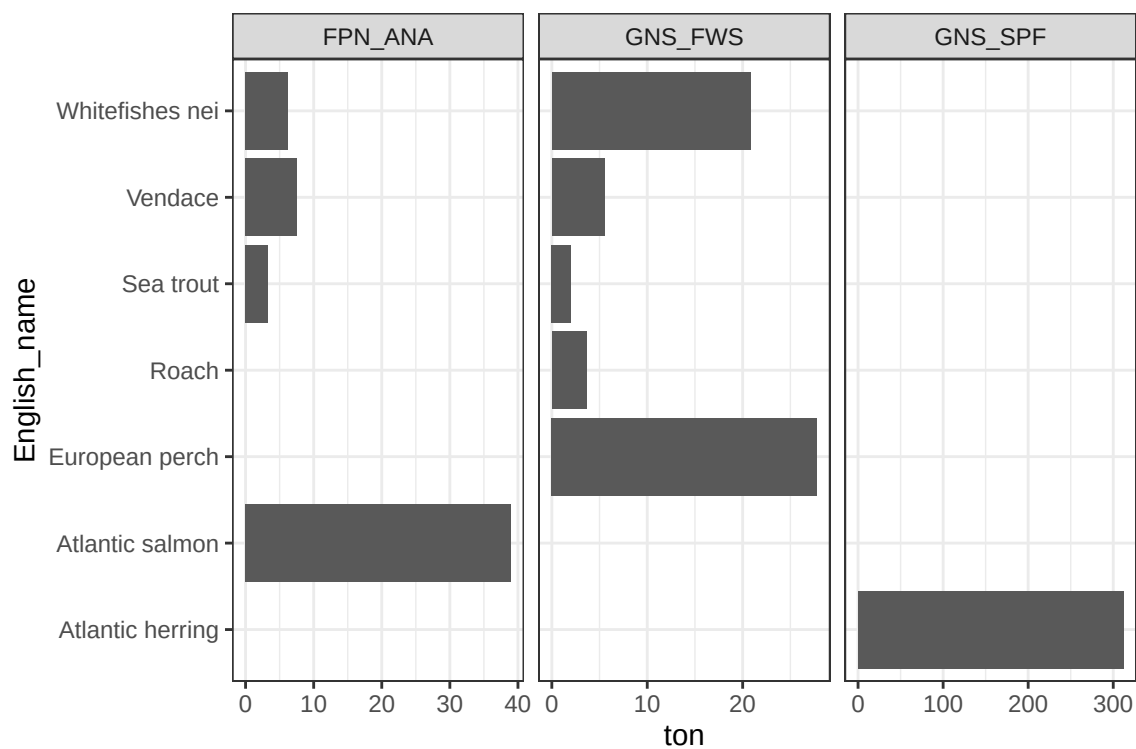


Figure C.4: Catches by metier, swedish small scale coastal fleet in Bothnian Sea (selection of metiers)

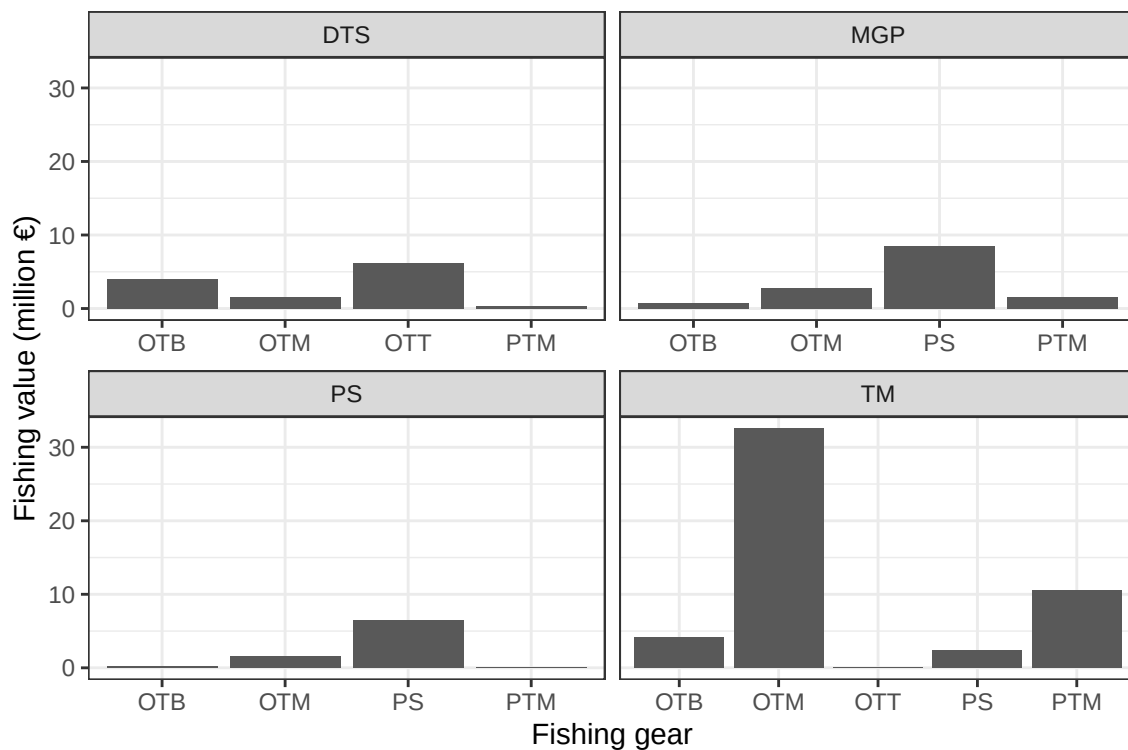


Figure C.5: Catches by gear and fishing method, swedish large scale fleet

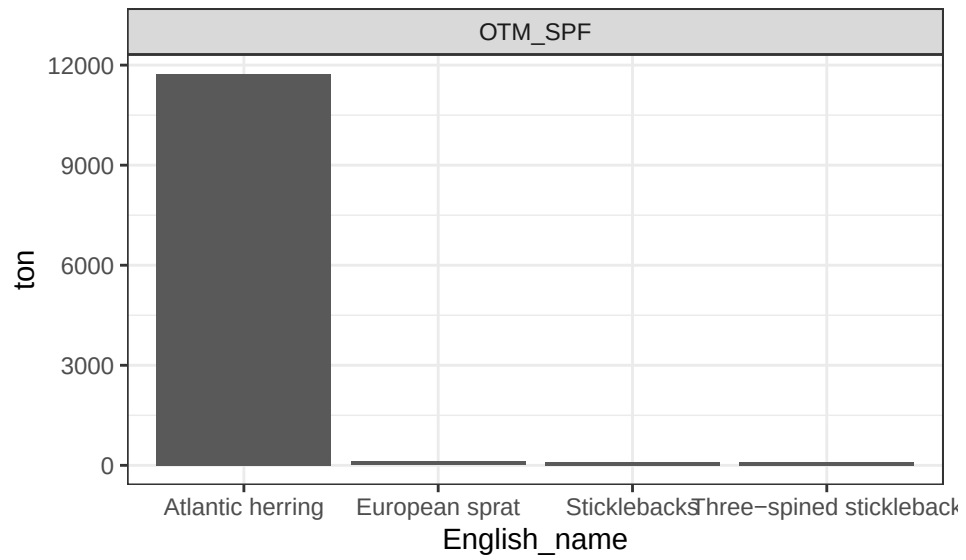


Figure C.6: Catches for pelagic trawlers, swedish fleet in Bothnian Sea

Recreational fishers

For recreational fishers we are interested in the distribution of number of practitioners by target species.

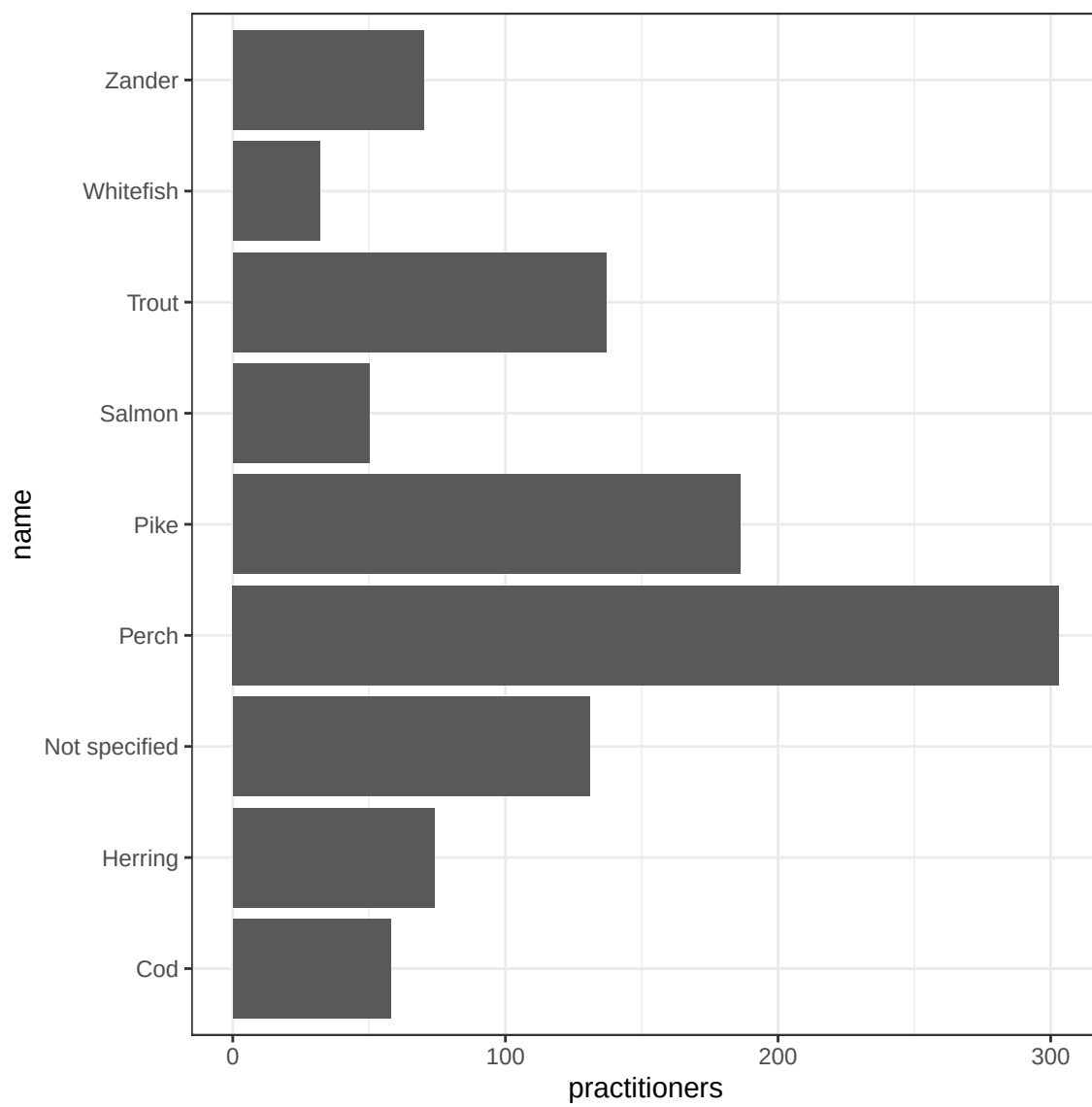


Figure C.7: Catches for recreational fishers

Economic data

Methods

The cost node reflects the economic impact of one single extreme weather event compared to annual average expenses. Cost categories to be used in the conditional probability table are: *negligible* (cost < 0.05%); *low* (0.05% < cost < 2.5%); *medium* (2.5% < cost < 5%); *high* (> 5%).

The conditional probability tables are estimated through the function `cost.cpt`, here below reported. The input needed are:

- expenses (mu and sd): annual total expenses, including maintenance, fuel, personnel. This value is anchored in literature, when available;
- fuel.cost (mu and sd): average cost for single fishing trip. This value is anchored in literature, when available;
- damage.small (lo and hi): self declared cost for repairing a minor damage;
- damage.large (lo and hi): self declared cost for repairing a large damage;
- damage.unc: self declared uncertainty on the cost for repairing damages;
- fuel.avg (lo and hi): self declared fuel consumption for single average fishing trip;
- fuel.large (lo and hi): self declared fuel consumption for single longer-than-average fishing trip;
- miss.catch (mu and sd): average gross economic value of catches from one fishing day
- economic.unc: self declared uncertainty associated with answer regarding expenses;
- fishing style: label describing the fishing style.

```
cost.cpt=function(expenses.mu, expenses.sd,
                  fuel.cost.mu, fuel.cost.sd,
                  damage.small.lo, damage.small.hi, damage.large.lo, damage.large.hi,
                  damage.unc,
                  fuel.avg.lo, fuel.avg.hi,
                  fuel.large.lo, fuel.large.hi,
                  miss.catch.mu, miss.catch.sd,
                  economic.unc,
                  f.style){

  # creating distributions
  ## distribution for annual expenses
  cost.annual=rtruncnorm(a=0, b=Inf,
                        mean=expenses.mu,
                        sd=expenses.sd, n=1000)

  ## distribution for minor damage
  cost.lo=rtruncnorm(a=damage.small.lo, b=damage.small.hi,
                    mean=(damage.small.hi-damage.small.lo)/2,
                    sd=damage.small.hi*damage.unc, n=1000)

  ## distribution for major damage
  cost.hi=rtruncnorm(a=damage.large.lo, b=damage.large.hi,
```

```

        mean=(damage.large.hi-damage.large.lo)/2,
        sd=damage.large.hi*damage.unc, n=1000)

## Distribution for avg travel Unit does not matter as long as it is the same of distribution for lar
avg.fuel=(fuel.avg.hi+fuel.avg.lo)/2
fuel.lo=rtruncnorm(a=fuel.avg.lo, b=fuel.avg.hi,
                  mean=avg.fuel,
                  sd=avg.fuel*economic.unc, n=1000)

## Distribution for large travel. Unit does not matter as long as it is the same of distribution for
lrg.fuel=(fuel.large.hi+fuel.large.lo)/2
fuel.hi=rtruncnorm(a=fuel.large.lo, b=fuel.large.hi,
                  mean=lrg.fuel,
                  sd=lrg.fuel*economic.unc, n=1000)

# Cost for fuel in extra travel. Multiplies proportion of large to avg travel by monetary cost.
extra.fuel.cost=(fuel.hi/fuel.lo)*rtruncnorm(a=0, b=Inf,
                                           mean=fuel.cost.mu,
                                           sd=fuel.cost.sd, n=1000)

# missing catches
miss.catch=rtruncnorm(a=0, b=Inf, mean=miss.catch.mu, sd=miss.catch.sd, n=1000)

# create data frame
cost.df=data.frame(minor.no=cost.lo, major.no=cost.hi, no.travel=extra.fuel.cost)
cost.df$minor.travel=cost.df$minor.no+cost.df$no.travel
cost.df$major.travel=cost.df$major.no+cost.df$no.travel
cost.df.2=cost.df+miss.catch
names(cost.df.2)=paste(names(cost.df.2), 'miss', sep='.')
cost.df.2$no.no.miss=miss.catch
cost.df=cbind(cost.df, cost.df.2)
cost.df[,]=apply(cost.df[,],2,function(x) x/cost.annual)
cost.df.disc=cost.df
cost.df.disc[,]=apply(cost.df.disc[,],2,
                     function(x) cut(x, c(-Inf, 0.005,0.025,0.05,Inf),
                                     labels=c('negligible','low','medium','high'))))

cost.df.disc$fishing_style=f.style
cost.cpt=cost.df.disc%>%
pivot_longer(-fishing_style, names_to = 'parent', values_to = 'state')%>%
dplyr::group_by(fishing_style,parent, state)%>%
tally()%>%
dplyr::mutate(prob=n/sum(n))

return(cost.cpt)
}

error.propagation=function(x.mu, x.er, y.mu, y.er){
  z.mu=x.mu/y.mu
  z.er=sqrt(((x.er/x.mu)^2)+((y.er/y.mu)^2))*z.mu
  return(data.frame(mu=z.mu, sd=z.er))
}

```

Small scale fisherman

Economic information for small scale coastal fishers is available in dataset D3 (Table C.1), are filtered for vessel length categories VL0010 and VL0812 and on fishing technologies FPO and DFN. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. Gross value of catches in a single fishing trip is obtained by dividing total annual value of catches by number of fishing trips. Other values are provided in the interviews.

Table C.1: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
2	minor_damage_cost	10000	50000	sek_unit
3	minor_damage_cost	1	50000	sek_year
7	minor_damage_cost	1000	50000	sek_unit
2	major_damage_cost	100000	500000	sek_unit
3	major_damage_cost	50000	200000	sek
7	major_damage_cost	50000	100000	sek_unit
3	fuel_trip	1500	2500	sek_day
3	extra_fuel	3000	5000	sek_day
7	extra_fuel	50	100	B liter_trip

Table C.2: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	1702	1567	thousand SEK	STECF_AER
Fuel cost	6670	8668	SEK	STECF_AER

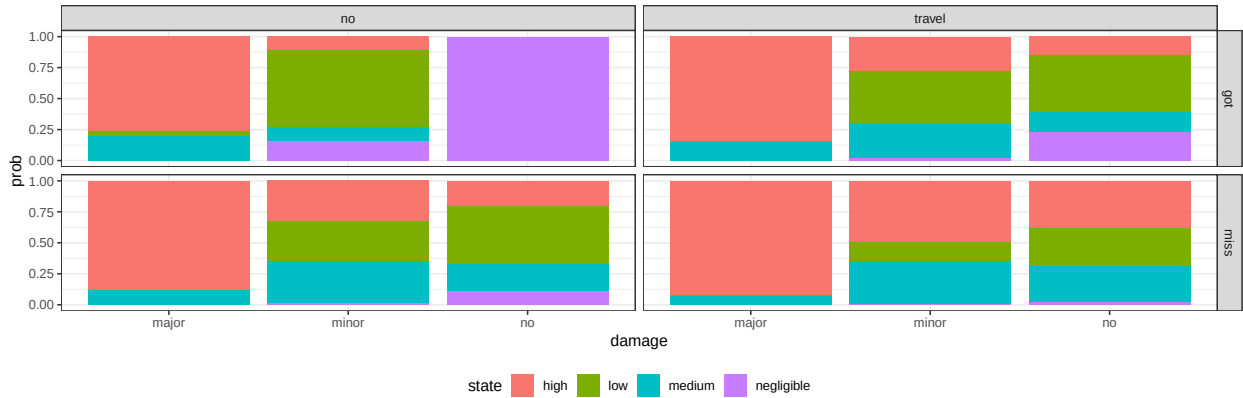


Figure C.8: CPT estimated for coastal fisherman

Pelagic trawlers

Pelagic economic information is derived from the STECF_AER data source (STECF 2024). STECF_AER data are filtered for vessel length categories VL2440 and VL40XX, including all fishing technologies. Cost of fuel per trip is calculated by dividing total fuel expenses by number of fishing trips. Annual spending is estimated by dividing total annual expenses by number of fishing vessels. The cost of longer travel was not available, and it is based on the assumption that an average travel costs between 80% and 120% of the average fuel cost, while a longer travel is between 100% and 300% of average fuel cost. Other values are provided in the interviews.

[1] 4981303 5279201 5837720 5749069 5529277 6694086

Table C.3: Variables from interviews

id_I	short_description	range.lwr	range.upr	unit_range
6	minor_damage_cost	10000	50000	sek_unit
6	major_damage_cost	100000	500000	sek_unit
6	fuel_trip	600	650	liters_hour
6	extra_fuel	NA	NA	not_reported

Table C.4: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	50404	5643	thousand SEK	STECF_AER
Fuel cost	259142	117662	SEK	STECF_AER

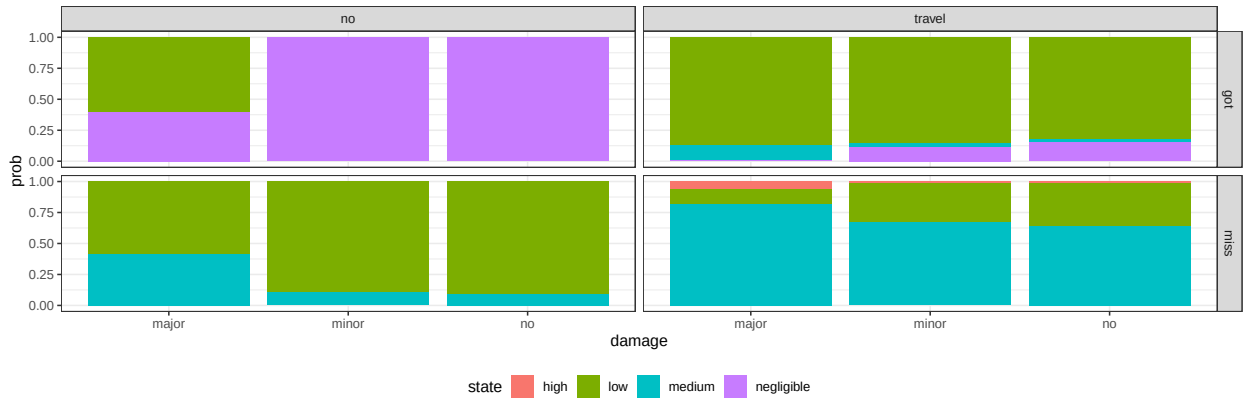


Figure C.9: CPT estimated for pelagic trawlers

Recreational

Pelagic economic information is derived from the recreational fisheries statistical report of the Swedish Department for Economic and Statistic Analysis (SWAM) data source (SWAM 2024). SWAM economic data represent the totality of recreational fisherman in Sweden, with no distinction between region or inland vs coastal fishing. Cost of fuel per trip is separately calculated for travel and for boat fuel. These two costs are then averaged. All the uncertainties provided in SWAM (on costs and on number of practitioners) are accounted for by propagating errors (REF). Annual spending is estimated by dividing total annual expenses by number of practitioners. Other values are provided in the interviews.

id_I	short_description	range.lwr	range.upr	unit_range
5	minor_damage_cost	10000	50000	sek_unit
5	major_damage_cost	50000	1000000	sek_unit
5	fuel_trip	30	50	B liter_trip
5	extra_fuel	20	50	B liter_trip

Table C.6: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	9213	3943	SEK	SWAM
Fuel cost	101	31	SEK	SWAM

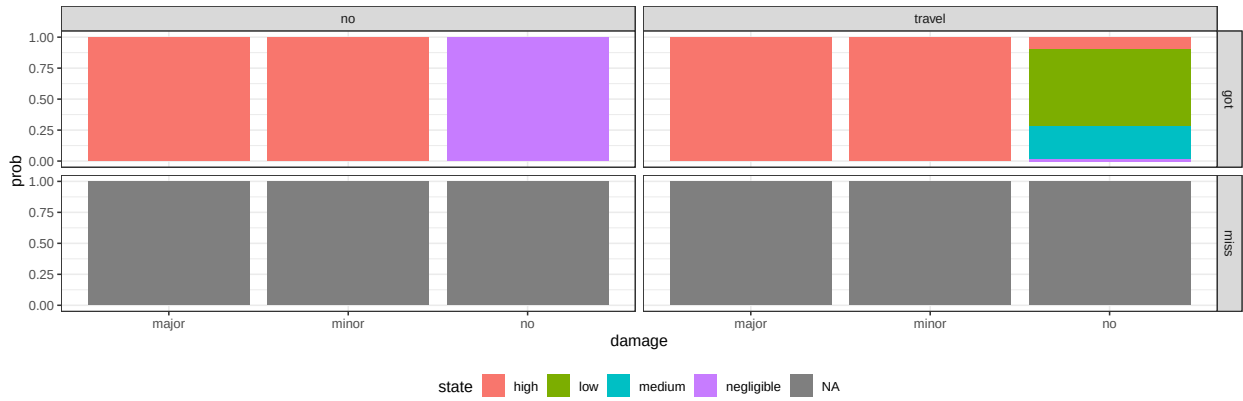


Figure C.10: CPT estimated for recreational fisherman

Fishing guide

No updated public information is available for fishing guide economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
1	season_maint	50000	10000	sek_season
1	minor_damage_cost	10000	20000	sek_unit
1	major_damage_cost	20000	6000000	sek_unit
1	fuel_trip	1000	1000	sek_day
1	extra_fuel	500	500	sek_day

Table C.8: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	175321	14244	SEK	Our estimation based on interviews

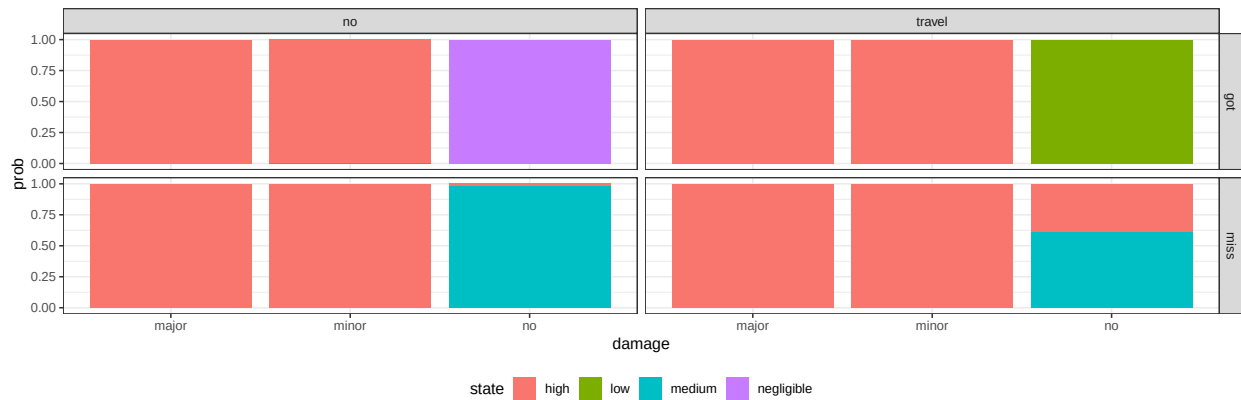


Figure C.11: CPT estimated for fishing guide

Household

No updated public information is available for archipelago fishers economic aspects in Sweden. All estimations base on interview results and our assumptions. Seasonal expenses are estimated by summing the declared maintenance costs and an estimation of fuel costs.

id_I	short_description	range.lwr	range.upr	unit_range
4	season_maint	NA	5000	sek_season
4	minor_damage_cost	200	2500	sek_unit
4	major_damage_cost	2500	5000	sek_unit
4	fuel_trip	1000	6000	sek_season
4	extra_fuel	NA	NA	not_reported

Table C.10: Variables from literature

variable	value_mean	value_sd	unit	source
Annual expenses	9037	1778	SEK	Our estimation based on interviews

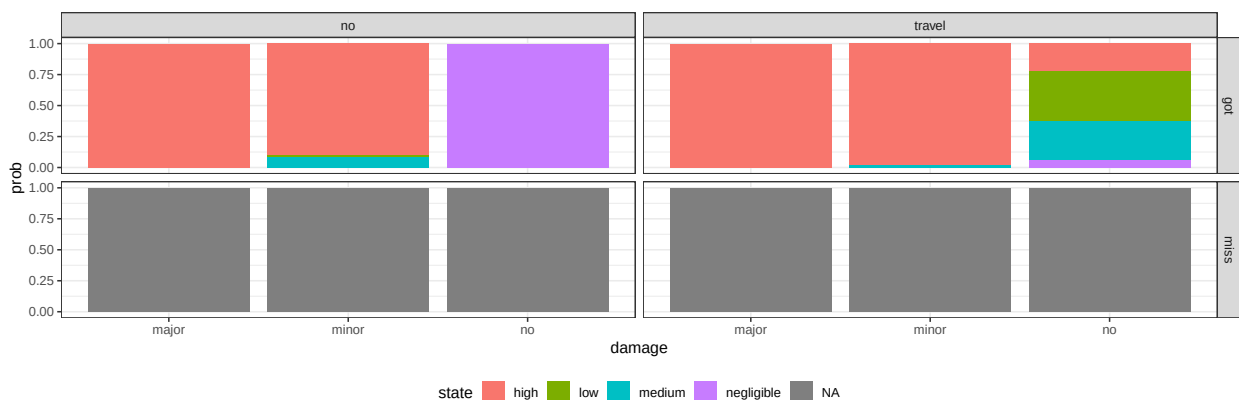


Figure C.12: CPT estimated for archipelago fisher

- FAO. 2026. “ASFIS List of Species for Fishery Statistics Purposes - All Information Collections.” <https://www.fao.org/fishery/en/collection/asfis>.
- JRC. 2025. “Scientific Technical and Economic Committee for Fisheries (STECF) – The 2024 Aquaculture Economic Report (STECF 24-14).” <https://publications.jrc.ec.europa.eu/repository/handle/JRC140767>.
- . n.d. “2025 - Fisheries Effort, Landings, Catches, Capacity, Biological Data up to 2024.” <https://doi.org/10.2905/JRC.KJFN63G>.
- STECF. 2024. *The 2024 Annual Economic Report on the EU Fishing Fleet (STECF 24-03 & 24-07)*. Luxembourg: Publications Office of the European Union. <https://publications.jrc.ec.europa.eu/repository/handle/JRC139642>.
- SWAM. 2024. “Fritidsfiske 2024.”